

OrderPulse — Step 6: Next.js Frontend Application

Technical Instruction & Execution Blueprint for AI Developer Agents

OBJECTIVE

Build the Next.js frontend application featuring a customer checkout interface, real-time stock indicators, and an admin analytics dashboard listening to real-time order updates.

Step 6 Tasks

- **1. App Initialization:** Initialize Next.js project with TypeScript, Tailwind CSS, and App Router inside frontend/.
- **2. API Handlers:** Create Route Handlers bridging Next.js to backend Go microservice endpoints.
- **3. Storefront UI:** Build interactive product catalog and order placement form with instant toast feedback.
- **4. Admin Dashboard:** Build live analytics feed and job status monitoring page.
- **5. Real-Time Integration:** Implement Server-Sent Events (SSE) or WebSockets to stream background processing updates straight to the UI.

Directory Layout

```
orderpulse/
├── frontend/
│   ├── app/
│   │   ├── api/
│   │   │   ├── inventory/route.ts
│   │   │   └── orders/route.ts
│   │   ├── dashboard/
│   │   │   └── page.tsx
│   │   ├── layout.tsx
│   │   └── page.tsx
│   ├── components/
│   │   ├── InventoryCard.tsx
│   │   ├── OrderForm.tsx
│   │   └── RealtimeFeed.tsx
│   └── package.json
```

Verification Commands

```
# Install dependencies
cd frontend && npm install

# Start development server
npm run dev

# Open UI at http://localhost:3000
```